DRAFT — P1–P5 MEASURED; P4-a-block-energy HIT (excess_final=0.08727119221832025); P5 2HIT/4MISS; L-cool-dir INSUFFICIENT

A3 Experimental Appendix

Configs, seeds, plot refs, and raw measurements for the official report.
Markdown only (report support) — not a separate study PDF.

Environment

Field	Value
Repo branch	`cursor/a3-from-desktop` (merged `a3-optimization2-81fe` @ `958f46c`)
Starter SHA	course `a3_release` `57c60bb`
Modal env	`cs312-daxton` (Mac / course auth)
W&B	`daxgute-stanford-university` / `assignments` (**MEASURED** from Batch 1 prefix)
Study PDF policy	report + intuition guide (+ optional quiz); no appendix PDF

P1 Mode Connectivity

Config

Key	Value
`lr_schedule`	`wsd0.0`
PREFIX_STEPS xx	`{0, 50, 100, 250, 500, 1000}`
BRANCH_STEPS yy	`{25, 50, 100, 250, 500}`
`batch_order_seed`	`{1, 2}`
EXPERIMENT_KEY	`a3-mode-connectivity`
peak LR	`0.003`
seed (model/data)	`model_seed=42`, `data_seed=42`
Checkout	`~/Desktop/assignments` @ `cursor/a3-from-desktop` `958f46c`

Runs

Stage	Run name / ID	Status	Notes
prefix	`model-d8-lr0.003-tok614M-wsd0.0-stop1000` · W&B `rq1k0bz1` · Modal `ap-L6RKrCcPCbhhgBvr3DLJfU`	**DONE**	end `val_loss=3.48832`
branches	12 runs · Modal `ap-57b3QORBZcKneFVkuuq7F3`	**DONE**	all keep-ckpts on volume
measure	Modal `ap-ZU7fsZIl4Mq2Wi5veNzXCb`	**DONE**	JSON below

Artifacts

Artifact	Path
Raw measure JSON (repo)	`experiments/a3_optimization/results/a3-mode-connectivity.json` (20091 B, 2026-10-10 17:21 PDT)
Store copy	`media/p1/a3-mode-connectivity.json`
Barrier table CSV	`media/p1/barrier-table.csv`
Summary JSON	`media/p1/barrier-summary.json`

Plots

Figure	Content	Path	Label
P1-b heatmap	Annotated (B(x,y))	`media/p1/plots/B_heatmap.png`	**MEASURED**
P1 Fit A	(B) vs (y) (+ FITTED lines)	`media/p1/plots/B_vs_y_curves.png`	MEASURED points + **FITTED**
P1 Fit B	(B) vs (x) @ (y=500)	`media/p1/plots/B_vs_x_y500.png`	MEASURED points + **FITTED**
P1-a / Fit C	(L_{\mathrm{ens}}) vs (L_{\mathrm{avg}}); (G) vs (B)	`media/p1/plots/ensemble_vs_avg.png`	MEASURED points + **FITTED**

Barrier numbers: `media/p1/barrier-table.csv` · `media/p1/barrier-summary.json`.
FITTED equations (Fits A/B/C): `internal/p1-scaling-fit-status.md` · intuition-guide Appendix Q1.1. **Never label FITTED as MEASURED.**

Barrier summary (MEASURED)

B=L(θavg)−12(L1+L2)B = L(\theta_{\mathrm{avg}}) - \tfrac12(L_1+L_2) at α=0.5\alpha=0.5.
Source: `media/p1/barrier-summary.json` (derived from `a3-mode-connectivity.json`).

Stat	Value
Cells	30
BminB_{\min}	−0.1161 at (x,y)=(250,100)(x,y)=(250,100)
BmaxB_{\max}	0.6237 at (0,500)(0,500)
BmeanB_{\mathrm{mean}}	−0.0334
B≤0.01B\le 0.01	**26 / 30**
Failures (B>0.01B>0.01)	(0,250)(0,250), (0,500)(0,500), (50,500)(50,500), (100,500)(100,500)

BB heatmap (MEASURED; ✓ = B≤0.01B\le 0.01)

xx \ yy	25	50	100	250	500
0	−0.0288 ✓	−0.0416 ✓	−0.0275 ✓	0.2173	0.6237
50	−0.1009 ✓	−0.1053 ✓	−0.0910 ✓	0.0055 ✓	0.1299
100	−0.0932 ✓	−0.1019 ✓	−0.0972 ✓	−0.0512 ✓	0.0319
250	−0.0963 ✓	−0.1106 ✓	−0.1161 ✓	−0.0779 ✓	−0.0522 ✓
500	−0.0747 ✓	−0.0915 ✓	−0.0953 ✓	−0.0910 ✓	−0.0763 ✓
1000	−0.0597 ✓	−0.0769 ✓	−0.0855 ✓	−0.0855 ✓	−0.0825 ✓

Measured cells — endpoints, ensemble, midpoint, BB (MEASURED; ✓ means B≤0.01B\le 0.01)

x	y	L(θ₁)	L(θ₂)	L_ens	L(θ_avg)	B
0	25	6.4380	6.4653	6.4077	6.4228	-0.0288 ✓
0	50	5.6329	5.6864	5.5798	5.6180	-0.0416 ✓
0	100	5.0282	5.0396	4.9238	5.0064	-0.0275 ✓
0	250	4.3202	4.3092	4.2046	4.5319	0.2173
0	500	3.7511	3.7483	3.6454	4.3734	0.6237
50	25	5.2708	5.2877	5.1962	5.1784	-0.1009 ✓
50	50	5.0453	5.0449	4.9441	4.9398	-0.1053 ✓
50	100	4.6943	4.6940	4.5872	4.6031	-0.0910 ✓
50	250	4.1278	4.1245	4.0231	4.1316	0.0055 ✓
50	500	3.7009	3.6898	3.5943	3.8253	0.1299
100	25	4.8268	4.8219	4.7374	4.7311	-0.0932 ✓
100	50	4.6870	4.6719	4.5790	4.5776	-0.1019 ✓
100	100	4.4887	4.4439	4.3591	4.3691	-0.0972 ✓
100	250	3.9961	3.9747	3.8808	3.9342	-0.0512 ✓
100	500	3.6670	3.6582	3.5634	3.6945	0.0319
250	25	4.2400	4.2449	4.1581	4.1461	-0.0963 ✓
250	50	4.1383	4.1451	4.0505	4.0311	-0.1106 ✓
250	100	3.9972	3.9767	3.8869	3.8709	-0.1161 ✓
250	250	3.7429	3.7357	3.6447	3.6615	-0.0779 ✓
250	500	3.5740	3.5712	3.4844	3.5204	-0.0522 ✓
500	25	3.7266	3.7154	3.6522	3.6463	-0.0747 ✓
500	50	3.7105	3.6971	3.6226	3.6123	-0.0915 ✓
500	100	3.6654	3.6611	3.5798	3.5680	-0.0953 ✓
500	250	3.5767	3.5748	3.4918	3.4847	-0.0910 ✓
500	500	3.4862	3.4857	3.4037	3.4097	-0.0763 ✓
1000	25	3.4786	3.4796	3.4221	3.4194	-0.0597 ✓
1000	50	3.4730	3.4757	3.4051	3.3974	-0.0769 ✓
1000	100	3.4619	3.4611	3.3867	3.3760	-0.0855 ✓
1000	250	3.4241	3.4225	3.3449	3.3378	-0.0855 ✓
1000	500	3.3812	3.3810	3.3017	3.2986	-0.0825 ✓

P2 River-Valley

Config

Key	Value
schedules	`wsd0.0`, `linear`, `cos`, `wsd0.2`
`ema_decays`	`(0.99, 0.999)` on schedule arms
`lr_dip`	on one `wsd0.2` arm (`dip0.4–0.45–0.5x0.1`)
horizon-free	`invsqrt`, `sgdr2000`
Batch	Batch 2 · Modal `ap-Izg7CMIJgyf29zQokn4SYv`

Batch 2 jobs (MEASURED 9/9)

Finals from `media/p2/batch2_finals.csv` (`state=finished`, `progress=1.0`). P3 sharpness from `media/p2/p3_sgd_sharpness_summary.json` / `media/p2/p3_adamw_sharpness_summary.json` only — blanks mean not in those sources (do not invent).

#	label	run_name	W&B	state	Label	train_loss	val_loss	op
0	a3-b2-p2-wsd0.0	model-d8-lr0.003-tok614M-wsd0.0-ema0.99-0.999	widt420l	finished	**MEASURED**	3.0593721866607666	3.0975050926208496	93
1	a3-b2-p2-linear	model-d8-lr0.003-tok614M-ema0.99-0.999	r0294p93	finished	**MEASURED**	2.886288642883301	2.9282028675079346	93
2	a3-b2-p2-cos	model-d8-lr0.003-tok614M-cos-ema0.99-0.999	sktw67q8	finished	**MEASURED**	2.8933908939361572	2.934119462966919	93
3	a3-b2-p2-wsd0.2	model-d8-lr0.003-tok614M-wsd0.2-ema0.99-0.999	soosjcxr	finished	**MEASURED**	2.8900909423828125	2.93774151802063	93
4	a3-b2-p2-wsd0.2-dip	model-d8-lr0.003-tok614M-wsd0.2-dip0.4-0.45-0.5x0.1	rms7iyyt	finished	**MEASURED**	2.8953793048858643	2.938987970352173	93
5	a3-b2-p2-invsqrt	model-d8-lr0.003-tok614M-invsqrt-warmupsteps94	1jri34em	finished	**MEASURED**	2.983588457107544	3.020456075668335	93
6	a3-b2-p2-sgdr2000	model-d8-lr0.003-tok614M-sgdr2000-warmupsteps94-minlr0.1	51b3qq4n	finished	**MEASURED**	2.996020793914795	3.0348048210144043	93
7	a3-b2-p3-0-sgd	model-d8-lr0.01-bs32-epochs1500.0-tok32.8K-wd0.0-emb16-nogradclip-b10.0-sgd-constant	0dewkykj	finished	**MEASURED**	1.268615484237671	8.053705215454102	14
8	a3-b2-p3-1-adamw	model-d8-lr0.003-epochs0.32-tok614M-warmup0.03125-wsd0.0	foew12y0	finished	**MEASURED**	3.2420499324798584	3.2397050857543945	29

Artifacts: `media/p2/batch2_finals.csv`, `media/p2/batch2_final_metrics.json`, `media/p2/p3_sgd_sharpness_summary.json`, `media/p2/p3_adamw_sharpness_summary.json`, `media/p2/p3_sharpness_medians.csv`

P3 Edge of Stability

Config

Key	Value
Starter core	full-batch SGD + minibatch AdamW sharpness (2 jobs in Batch 2)
Batch	Batch 2 · Modal `ap-Izg7CMIJgyf29zQokn4SYv`

Runs (MEASURED finals + sharpness)

Jobs 7–8 from the Batch 2 table above. Sharpness columns only:

#	label	W&B	Label	train_loss	val_loss	step	med. (\lambda_{\max})	med. (\e (\eta\lar
7	a3-b2-p3-0-sgd	0dewkykj	**MEASURED**	1.268615484237671	8.053705215454102	1499	245.16314737956807	2.4516314
8	a3-b2-p3-1-adamw	foew12y0	**MEASURED**	3.2420499324798584	3.2397050857543945	2999	10329.238638171333	30.987715

Artifacts: `media/p2/batch2_finals.csv`, `media/p2/p3_*_sharpness_summary.json` (twins under `media/p3/`)

P4 Inside the Hessian

Label rule: fill P4 value cells only from store JSON (`media/p4/…` / `hessian_jobs.load_result`). Stage a is **13/13 COMPLETE** — all `a-adamw-*` steps are **MEASURED**. Do **not** invent (\lambda) / density / subspace numbers. Handout bands (\lambda_{\max}\in[4.5,6.0]), val (\in[2.925,2.931]) are assignment-given, not our MEASURED. Ledger `P4-a-*` HIT/MISS from `internal/p4-9375-hit-miss-summary.md` (operator copy of dry-run DRAFT_*); `P4-a-block-energy` **HIT** (MEASURED excess_final=0.08727119221832025; fam_above=True; layer_above=True from `a-random-energy-1526/9375`). Stages b/c scored: `P4-invar-dL` / `P4-top10-frac` / `P4-c-loss-rank` **MISS** (see dry-runs).

Config

Key	Value
Batch	Batch 3 · Modal stage `a` `ap-DAUDZsFRech705uMDN7q3x` (**13/13** volume; `a-adamw-{0,3,15,61,153,305,763,1526,3052,4578,6104,7630,9375}` in store)
Probe rows	64
Precision	FP32, TF32 off
Provided ckpts	shared volume `hard-dl-dclm-v1` → `/root/shared_data/a3_hessian/runs/`
AdamW measure steps (`STEPS`)	`0, 3, 15, 61, 153, 305, 763, 1526, 3052, 4578, 6104, 7630, 9375`
Density at	`(0, 153, 1526, 9375)`
Families	`embedding`, `attention`, `qk_norm`, `mlp`, `norm`, `head`
(c) continue	SGDM @ 1526; (k=10); 300 steps; refresh 25; arms `full` / `top` / `removed`

Stage status

Stage	Job pattern	Count (starter)	Status	Result path
`a`	`a-adamw-{step}`	13	**DONE** 13/13	`media/p4/stage-a/a-adamw-{0,3,15,61,153,305,763,1526,3052,4578,6104,7630,9375}.json` (+ summaries)
`b-rescale`	`b-rescale-reference` + `b-rescale-head-x0.25.json`	≥2	**DONE** — `P4-invar-dL` **MISS**	`media/p4/` + `/cursor/stores/bc-01947f99-cdd8-4c04-97a8-01a1028bf8af/media/p4/b-rescale-head-x0.25.json`
`b-train`	`TrainConfig` arms (freeze before launch)	TBD	**PENDING**	
`b-measure`	`b-final-{run}`	TBD	**PENDING**	
`c`	`c-full`, `c-top`, `c-removed`	3	**DONE**	`media/p4/stage-c/`
`fetch`	downloads finished JSON	—	**DONE** — stage a + b-rescale heads (x0.25/x0.5/x2) + c 3/3 in `media/p4/`	`results/` → store `media/p4/`

P4(a) — AdamW checkpoint measure (PARTIAL 13/13)

MEASURED rows from `media/p4/stage-a/` JSON/summary only. Ledger `P4-a-*` stays unresolved (**INSUFFICIENT**) until mid/final stage-a jobs land.

Job	step	Label	(\lambda_{\max})	(\lambda_{\min})	density?	Notes
`a-adamw-0`	0	**MEASURED**	1431.5030691763686	-1410.061273996107	yes	tokens=0; probe_loss=8.795355558395
`a-adamw-3`	3	**MEASURED**	1571.7420262270407	-601.6596057825041		tokens=196608; probe_loss=8.39551979303
`a-adamw-15`	15	**MEASURED**	53.36426571742344	-17.452665724452103		tokens=983040; probe_loss=6.92180022597
`a-adamw-61`	61	**MEASURED**	18.054717868295125	-1.8957655982714006		tokens=3997696; probe_loss=5.40091693401
`a-adamw-153`	153	**MEASURED**	6.502395300121604	-0.2562527330248557	yes	tokens=10027008; probe_loss=4.607155263423
`a-adamw-305`	305	**MEASURED**	4.32926630998023	-0.06854534676741643		tokens=19988480; probe_loss=4.108570218088
`a-adamw-763`	763	**MEASURED**	3.02398152405625	-0.08332850048247277		tokens=50003968; probe_loss=3.574245914816
`a-adamw-1526`	1526	**MEASURED**	2.1954286466134665	-0.0637674257861686	yes	tokens=100007936; probe_loss=3.378823339938
`a-adamw-3052`	3052	**MEASURED**	3.1637529312879873	-0.06470202230389413		tokens=200015872; probe_loss=3.220080047848
`a-adamw-4578`	4578	**MEASURED**	3.7423705126158477	-0.04897616951626389		tokens=300023808; probe_loss=3.134724542498
`a-adamw-6104`	6104	**MEASURED**	4.155164815834522	-0.03642192924304736		tokens=400031744; probe_loss=3.062897324566
`a-adamw-7630`	7630	**MEASURED**	5.281516516770698	-0.049697684311825786		tokens=500039680; probe_loss=3.006814658644
`a-adamw-9375`	9375	**MEASURED**	5.967310256347902	-0.08023598879266304	yes	tokens=614400000; probe_loss=2.947444289922 final

P4(a) — Family (\lambda_{\max}) by step (PARTIAL; steps 0, 3, 15, 61, 153, 305, 763, 1526, 3052, 4578, 6104, 7630, 9375 MEASURED)

step	embedding	attention	qk_norm	mlp	norm
0	1414.2983976747726	9.878542601155342	0.006507140133876505	7.0111895050073985	0.019894242489
3	1553.9202300399772	11.031923073793246	0.007607915139353549	10.39684728301151	0.022457015802
15	45.81142761722067	3.5631538708123083	0.0052545383095276	4.881036613766535	0.047742727705
61	16.490874755734055	0.8209037679944203	0.01704810366362142	0.9108818181867759	0.027675075171
153	2.4572441518559858	0.3160409788135334	0.018457310532168113	0.31233264994292786	0.034257656270
305	0.43233615820632054	0.29590823510701564	0.021688786309741786	0.2235756892072057	0.063930591174
763	0.060253284132475486	0.42656153617737463	0.01967095423000681	0.12688991611937916	0.191194874445
1526	0.02298266769621421	0.35228002015156656	0.018872888134703898	0.11421332384774996	0.419747262683
3052	0.015344048212154827	0.4595499107430324	0.019440740722556856	0.18380350452107147	0.706210144370
4578	0.015767613208961386	0.5082573113267944	0.024574157359865772	0.24068523075468984	0.601541092020
6104	0.016467160202990005	0.6584555395717259	0.030617703609270873	0.3553282596782511	0.522832764614
7630	0.030232045836065136	0.9592068563649059	0.044463133646784565	0.5427710855233483	0.471010604025
9375	0.03250318241691776	1.2078980133022608	0.0592233945185601	0.7639929133496728	0.474576536122

(Other `STEPS` rows may be appended after fetch; do not invent. Families from full `media/p4/stage-a/a-adamw-*.json` — summaries may omit families.)

P4(b) — Rescale / train measure (MEASURED — `P4-invar-dL` MISS)

Baseline + after-arm from dry-run (`internal/p4-b-score-dryrun.md`). `P4-invar-dL` **MISS** — |ΔL| =2.73853462934494; λ ratio=0.4658784038724411; λ 5.967310256347902→2.780040977639008; probe_loss 2.9474442899227142→5.685978919267654.

Job	Arm	Label	(\lambda_{\max})	probe_loss	(\lvert\Delta L_{\mathrm{val}}\rvert)	(\lambda) rat ref
`b-rescale-reference`	reference @ 9375	**MEASURED**	5.967310256347902	2.9474442899227142	—	— (baseline)
`b-rescale-head-x0.25`	hand rescale (after-arm)	**MEASURED**	2.780040977639008	5.685978919267654	2.73853462934494	0.46587840387
`b-rescale-head-x2`	hand rescale (optional)	**MEASURED**	7.302746614488089	3.9412150979042053	—	—
	train arm (TBD)	**PENDING**				

P4(c) — SGDM top subspace (MEASURED)

Job	transform	Label	end train/val	mean (|P_k g|/|g|)	Notes
`c-full`	full	**MEASURED**	3.582527816295624	0.09296940652964016	mean frac **MISS**
`c-top`	top	**MEASURED**	5.502330541610718	0.16483314082647363	
`c-removed`	removed	**MEASURED**	3.5821831226348877	0.027522441710655886	rank **MISS** gap=-0.000344693660736084

Plots

Figure	Description	Path	Status
P4-a early (\lambda_{\max})	Step vs (\lambda_{\max}) at all MEASURED STEPS 0…9375 (sorted)	`media/p4/p4_lambda_max_early.png`	**MEASURED** (13/13)
P4-a family (\lambda_{\max}) early	Step vs family (\lambda_{\max}) (embedding/attention/mlp/head) at all MEASURED STEPS 0…9375	`media/p4/p4_family_lambda_early.png`	**MEASURED** (13/13)
P4-a density	Eigenvalue density		**PENDING**
P4-c subspace	Loss / grad frac / (\lambda_{\max})		**PENDING**

P5 Optional

Field	Value
Question	Post-separation cooldown: same-valley walls vs true modes (design `docs/a3-p5-exploratory-design.md`)
Code diff path	`experiments/a3_optimization/p5_cooldown_barrier.py`
Resolving command	`uv run python -m experiments.a3_optimization.p5_cooldown_barrier measure`
Outcome	**MEASURED** — B_R=0.6237, B_C=0.7697049379348755, B_K=0.7320129871368408, G_C=0.8391239460323332, dB=-0.14600493793487546. Scores: `P5-B-cool-x0` **MISS** (B_C=0.7697049379348755); `P5-dB-cool` **MISS** (dB=-0.14600493793487546 (B_R=0.6237 [JSON] – B_C=0.7697049379348755)); `P5-B-const-ctrl` **HIT** (B_K=0.7320129871368408); `P5-rank-arms` **MISS** (B_C=0.7697049379348755 B_R=0.6237 B_K=0.732012987136840 8); `P5-G-cool-fitC` **HIT** (G_C=0.8391239460323332 B_C=0.7697049379348755

Canonical Batch 2 9/9 table (finals + P3 sharpness) lives under **P2 River-Valley → Batch 2 jobs** above. CSV: `media/p2/batch2_fi